NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) GTM- 1 Date: 18.12.2024
Time: 3 HRS JEE-MAIN Max. Marks: 300

KEYSHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	B	C	C	C	B	B	B	B	C
11	12	13	14	15	16	17	18	19	20
A	A	B	C	C	A	C	A	B	C
21	22	23	24	25					
6	150	20	12	6					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
A	C	A	D	B	D	A	C	C	B
36	37	38	39	40	41	42	43	44	45
A	B	B	B	B	D	A	C	C	C
46	47	48	49	50					
3	120	6	2	3					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	D	A	B	D	A	C	B	A	D
61	62	63	64	65	66	67	68	69	70
A	A	B	C	D	B	C	D	A	B
71	72	73	74	75					
7	3	8	6	5					

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HINTS&SOL

PHYSICS

1. Distance between two nodes $=\frac{\lambda}{2}=\frac{v}{2n}=\frac{16}{2n}=\frac{8}{n}$

2. Speed of sound in gases is given by $v=\sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{v_1}{v_2}=\sqrt{\frac{M_2}{M_1}}$

3. $\frac{1}{C'}=\frac{1}{C}+\frac{1}{C}+\frac{1}{C}=\frac{3}{C}$

$C'=\frac{C}{3}$

$V'=V_1+V_2+V_3=V+V+V=3V$

4. The expression represents energy density $=\frac{\text{Energy}}{\text{Volume}}$. So it's dimensions are

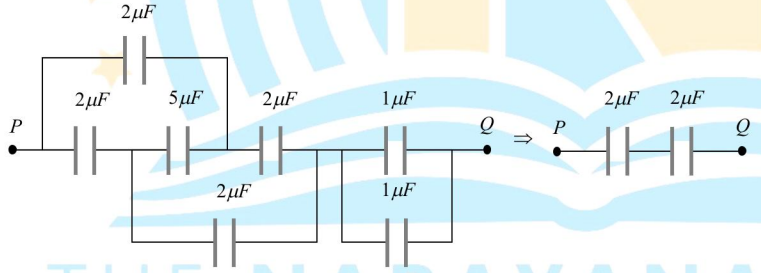
$\frac{ML^2T^{-2}}{L^3}=\left[ML^{-1}T^{-2}\right]$

5. Electrostatic potential energy $=\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{d}$

$\therefore \quad \text{Energy}=\frac{\left(9\times10^9\right)\times\left(+1.6\times10^{-9}\right)\times\left(-1.6\times10^{-19}\right)\text{J}}{10^{-10}}$

$=-9\times10^9\times1.6\times10^{-19}\times10^{10}\text{eV}=-14.4\text{eV}$

6. The given circuit can be redrawn as follows



$\Rightarrow C_{PQ}=1\mu\text{F}$

7. $P \propto V^2 \Rightarrow \frac{P}{P_0}=\left(\frac{V}{V_0}\right)^2 \Rightarrow P=\left(\frac{V}{V_0}\right)^2 P_0$

8. Here, V<E

$\therefore \quad E=V+Ir$

For first case, current is $\frac{12}{16}$A

$$E=12+\frac{12}{16}r \quad ....(i)$$

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For second case, current is $\frac{11}{10}$A

$$E = 11 + \frac{11}{10}r \quad .....(ii)$$

From Eqs. (i) and (ii),

$$12 + \frac{12}{16}r = 11 + \frac{11}{10}r$$

$$\Rightarrow r = \frac{20}{7}\Omega$$

9. Torque(τ) is given by $\tau = |M \times B|$

Where M is magnetic dipole moment of loop given by,

$M = iA$

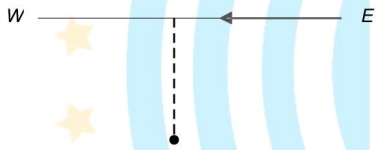
$\therefore M = (i\pi r^2)\hat{K}$

$\tau = |M \times B| = (\pi r^2 iB_x)$

Now, torque of weight will be mgr.

$$mgr = \pi r^2 iB_x \Rightarrow \quad i = \frac{mg}{\pi r B_x}$$

10. $B = \frac{\mu_0 I}{2\pi R}$

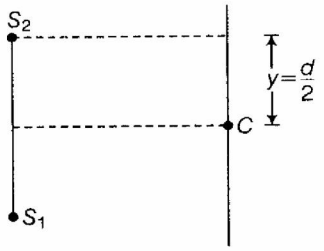


Direction is given by Right hand palm rule

$$B = \frac{4\pi \times 10^{-7} \times 100}{2\pi(4)}\text{T} = 5 \times 10^{-6}\text{T}$$ towards south

11. Here, path difference is

$$\Delta x = \frac{yd}{D} = \frac{d^2}{2D}$$



For first order minima,

$$\Delta x = \frac{\lambda}{2}$$

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Or $\frac{d^2}{2D}=\frac{\lambda}{2}$

$\therefore \quad \lambda=\frac{d^2}{D}$

12. Let a = acceleration of electron,
As a = constant
$\therefore$ E = uniform

$\Rightarrow s=ut+\frac{1}{2}at^2 \ (\because u=0) \Rightarrow a=\frac{2s}{t^2}$

Also, $a=\frac{\text{force}}{\text{mass}}=\frac{eE}{m}$

And $E=\frac{V}{s}$

So, $\frac{eV}{sm}=\frac{2s}{t^2}$

$\Rightarrow$ V= potential difference between plates.

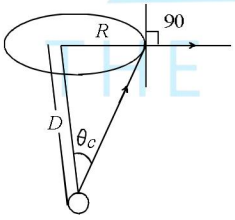
$\Rightarrow \frac{2s^2}{t^2}.\frac{m}{e}=\frac{2\times\left(2\times10^{-2}\right)^2}{\left(1.5\times10^{-8}\right)^2}\times\frac{9.1\times10^{-31}}{1.6\times10^{-19}}$

$=\frac{8\times9.1}{2.25\times1.6}=20.2\,V$

13. $\frac{\sin\theta_c}{\sin90}=\frac{1}{\mu}$

$\sin\theta_c=\frac{1}{\mu}$

$\Rightarrow \sin\theta_c=\frac{1}{n}$



$\Rightarrow \frac{R}{\sqrt{R^2+D^2}}=\frac{1}{n}\Rightarrow\frac{R^2+D^2}{R^2}=\frac{n^2}{1}$

$\Rightarrow 1+\frac{D^2}{R^2}=n^2-1\Rightarrow R^2=\frac{D^2}{n^2-1}$

$R=\frac{D}{\sqrt{n^2-1}}$

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If the bulb is not at seen through the surface R must be greater than

$$\frac{D}{\sqrt{n^2-1}}$$

14.

For two images for the two positions in displacement: $m_1 m_2 = 1$

So $O = \sqrt{A_1 A_2}$

15. $mv = \frac{h}{\lambda}$

or $v = \frac{h}{m\lambda} = \frac{6.6\times10^{-34}}{9.1\times10^{-31}\times5200\times10^{-10}} = 1.4\times10^{3}\, ms^{-1}$

16. $qvB = qE \Rightarrow v = \frac{E}{B}$

But $\frac{1}{2}mv^2 = qV$ so $\frac{q}{m} = \frac{v^2}{2V} = \frac{E^2}{2VB^2}$

17. The binding energy of nucleus may be defined as the energy equivalent to the mass defect of the nucleus.

If Δm is mass defect than according to Einstein's mass energy relation.

Binding Energy

$= \Delta mC^2 = \left[\left\{Zm_p + (A-Z)m_n\right\} - M\right]C^2$

$= (7\times1.00783 + 7\times1.00867 - 14.00307)C^2$

or $\mathrm{BE} = 0.1124\times931.5\,\mathrm{MeV}$

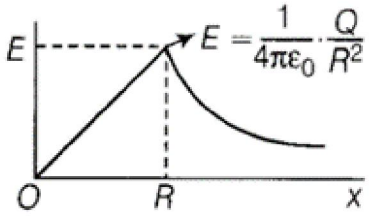
or BE = 104.6

18. B.E. per nucleon is maximum for Fe^{56}. For futher detail refer theory

19. All the three plates will produce electric field at point P along negative Z-axis

Hence, $\mathrm{E}_P = \left[\frac{\sigma}{2\varepsilon_0} + \frac{2\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0}\right](-\hat{\mathrm{k}}) = \frac{-2\sigma}{\varepsilon_0}\hat{\mathrm{k}}$

20.



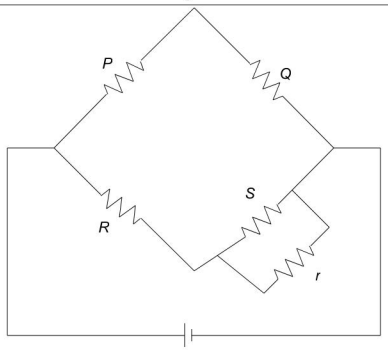
Thus, the field linearly with x, until x = R. It then decreases as $\frac{1}{X^2}$, as plotted in figure

21. Let a resistance r ohm be shunted with resistance S, so that the bridge is balanced.

If S' is the resultant resistance of S and r, then

In balanced position

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$$\frac{P}{Q}=\frac{R}{S'}$$

$$\frac{2}{2}=\frac{2}{S'}$$

$$\therefore S'=2\Omega$$

Now,

$$\frac{1}{S'}=\frac{1}{S}+\frac{1}{r}$$

$$\text{or}\frac{1}{r}=\frac{1}{S'}-\frac{1}{S}=\frac{1}{2}-\frac{1}{3}=\frac{3-2}{6}$$

$$\frac{1}{r}=\frac{1}{6}$$

$$r=6\Omega$$

22. For such sphere,

Center potential = 1.5 (Surface potential)

23. According to the problem, combination of L_1 and L_2 act a simple glass plate. Hence according to formula

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}-\frac{d}{f_1f_2}$$

$$\frac{1}{f_1}+\frac{1}{f_2}-\frac{d}{f_1f_2}=0\Rightarrow\frac{1}{f_1}+\frac{1}{f_2}=\frac{d}{f_1f_2}$$

$$\Rightarrow\frac{1}{30}-\frac{1}{10}=\frac{d}{30\times-10}\Rightarrow\frac{-20}{30\times10}=-\frac{d}{30\times10}\Rightarrow d=20cm$$

24. $L_D=v_o+u_e$ and for objective lens $\frac{1}{f_o}=\frac{1}{v_o}-\frac{1}{u_o}$

Putting the values with proper sign convention

$$\frac{1}{+2.5}=\frac{1}{v_o}-\frac{1}{(-3.75)}\Rightarrow v_o=7.5cm$$

For eye lens $\frac{1}{f_e}=\frac{1}{v_e}-\frac{1}{u_e}\Rightarrow\frac{1}{+5}=\frac{1}{(-25)}-\frac{1}{u_e}\Rightarrow u_e=-4.16cm$

$$\Rightarrow|u_e|=4.16cm$$

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Hence $L_D = 7.5 + 4.16 = 11.67\,cm$

25. $W = -M.B = -MB\cos\theta$

$\Delta W = W_f - W_i = -(-MB) - (-MB) = 2MB$

$= 2 \times (iAN) \times B = 2 \times 0.5 \times \pi \times (0.1)^2 \times 100 \times 2$

$= 2\pi J$

CHEMISTRY

26. $\lambda = \frac{12.25}{\sqrt{V}} \overset{0}{A}$

$\therefore$ V = 63.5 V

27. Mixtures of $CHCl_3$ and CH_3COCH_3 shows negative deviation from Raoult's law, so vapour pressure decreases and boiling point increases.

28. $Fe^{+3} + 3e^- \longrightarrow Fe$ $\quad \Delta G_1^0 = -nFE_1^0$

$Fe \longrightarrow Fe^{+2} + 2e^-$ $\quad \Delta G_2^0 = -nFE_2^0$

$Fe^{+3} + e^- \longrightarrow Fe^{+2}$ $\quad \Delta G_3^0 = -nFE_3^0$

$\Delta G_3^0 = \Delta G_1^0 + \Delta G_2^0$

$\Rightarrow -1 \times F \times E_3^0 = -3 \times F \times (-0.04) - 2 \times F \times (+0.44)$

$\therefore Fe^{+3} / Fe^{+2} = +0.76\,V$

29. Charge passed $= 0.1 \times 965$

$= 96.5$ C

$= \frac{96.5}{96500} F = 0.001F$

$\Rightarrow 10^{-3} F$ ---- 10^{-3} equivalents of Cu

---- $10^{-3} \times 63.5\,g$

---- 0.0635 g

$\therefore$ Current efficiency $= = \frac{0.03175}{0.0635} \times 100\%$

$= 50\%$

30. $K = \frac{1}{R}\left(\frac{\ell}{a}\right) \Rightarrow 0.0112 = \frac{1}{55}\left(\frac{\ell}{a}\right) \Rightarrow \frac{\ell}{a} = 0.616$

31. Slow step is the rate determining step (RDS) and (N_2O_2) is the reactive intermediate.

$\therefore r = k_2[N_2O_2][O_2]$ .... (i)

From reversible reaction, $[N_2O_2]$ is

$\frac{k_1}{k_{-1}} = \frac{[N_2O_2]}{[NO]^2}$ .... (ii)

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Substitute $[N_2O_2]$ in equation (i),

$$r = k_2\left(\frac{k_1}{k_{-1}}\right)[NO]^2[O_2]$$

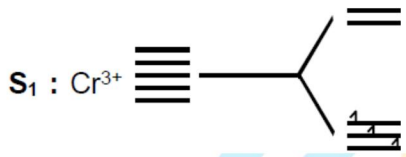
Hence, rate constant $= k_2\left(\frac{k_1}{k_{-1}}\right)$

32. $2KMnO_4 + 3H_2SO_4 \longrightarrow 2KHSO_4 + (MnO_3)_2SO_4 + 2H_2O$

$(MNO_3)_2SO_4 + H_2O \longrightarrow Mn_2O_7 + H_2SO_4$

$Mn_2O_7 \xrightarrow{\Delta} 2MnO_2 + \frac{3}{2}O_2$

34.



$CFSE = 3\times -0.4 = -1.2\Delta_0$, hybridization is d^2sp^3 (NH_3 is strong field ligand)

S_2 : $Fe^{3+}, 3d^5$ - one unpaired electron after pairing (CN^- is stronger field ligand)

$\therefore \mu = \sqrt{1\ (1+2)} \approx 1.73\,BM$

$S_3 : \left[\overset{II}{Fe}(\overset{-5}{CN})_5\overset{+}{NO}\right]^{2-}$ and $S_3 : \left[\overset{II}{Fe}(\overset{-5}{CN})_5 NOS^{-1}\right]^{4-}$

In reactant and product, the iron is in same oxidation state i.e. +2.

35. The order is based on the least stability of bromine oxide due to lesser polarity of X–O bond and larger size of overlapping orbitals as compared to iodine and chlorine respectively.

36. Poor shielding by f- and d-electrons enhances the effective nuclear charge in Bi. This causes contraction in size.

37. Fact from N.C.E.R.T

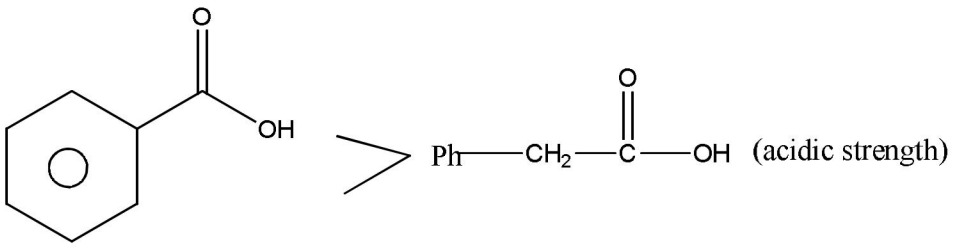
38. Alcohols are higher B.P than amines

$Et_2NH > Et_3N > EtNH_2 > NH_3$



is a primary amine

39. $CH_3 - CH_2 - COONa \longrightarrow CH_3 - CH_3$

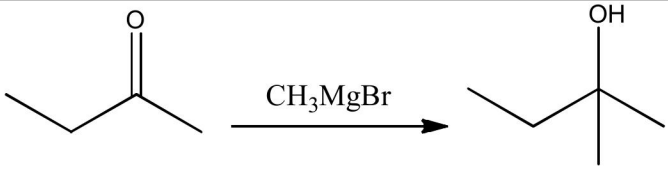


(acidic strength)

$H - C - OH \xrightarrow[2.H_2No\ O]{1.Cl_2/Rp}$ No reaction

40. Cellulose is a polymer of glucose. D.N.A is a polymer of nucleotide.

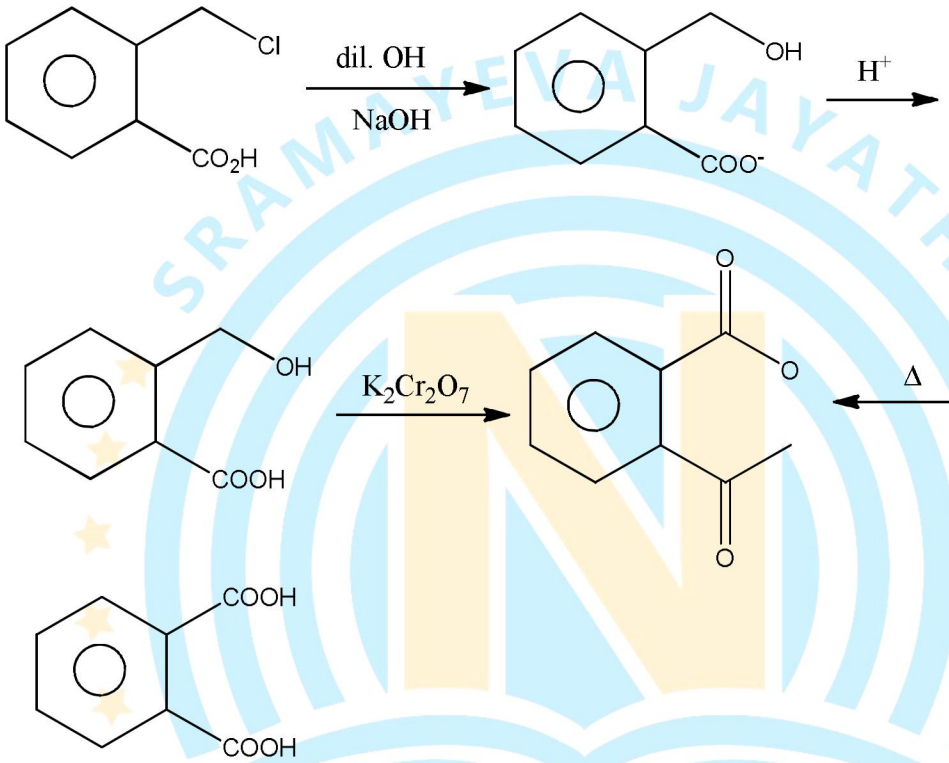
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41. 

42. - M & - I groups decreases acidity

43. Acetaldehyde is better acceptor than acetophenon

44. Aromatic I amines gives azodye test
Isopropyl alcohols gives iodoform test

45. 

46. For urea, $\Delta T_f = k_f \times m$ or $k_f = \frac{\Delta T_f}{m} = \frac{1.86}{1} = 1.86$

Now for CH_3COOH

$\Delta T_f = ik_f m$

So $I = 1.1$ Now $i = 1 + \alpha$ so $\alpha = 1.1 - 1 = 0.1$

Now	CH_3COOH	CH_3COO^-	+	H^+
	C	0		0
	$C - C\alpha$	$C\alpha$		$C\alpha$

$[H^+] = C\alpha = 0.01 \times 0.1 = 0.001$

So pH = 3.

47. Volume strength = $11.2\times$ Molarity

$K = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{30}$

For 1st order reaction

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$$\ln\frac{|A|}{|A|} = -kt$$

$$\Rightarrow \ln\frac{1}{16} = -\frac{\ln 2}{30}.t$$

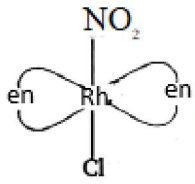
$$\Rightarrow t = +\frac{4\ln 2 \times 30}{\ln 2}$$

t = 120 minutes

48. $\left(Rh(en)_2Cl(NO_2)\right)$ and $\left[Rh(en)_2Cl(ONO)\right]$

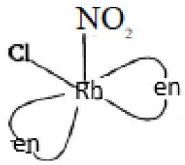
$\therefore$ two structural isomers

And each structural isomer has 3 stereo isomers each.

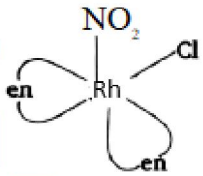


trans

Optically inactive

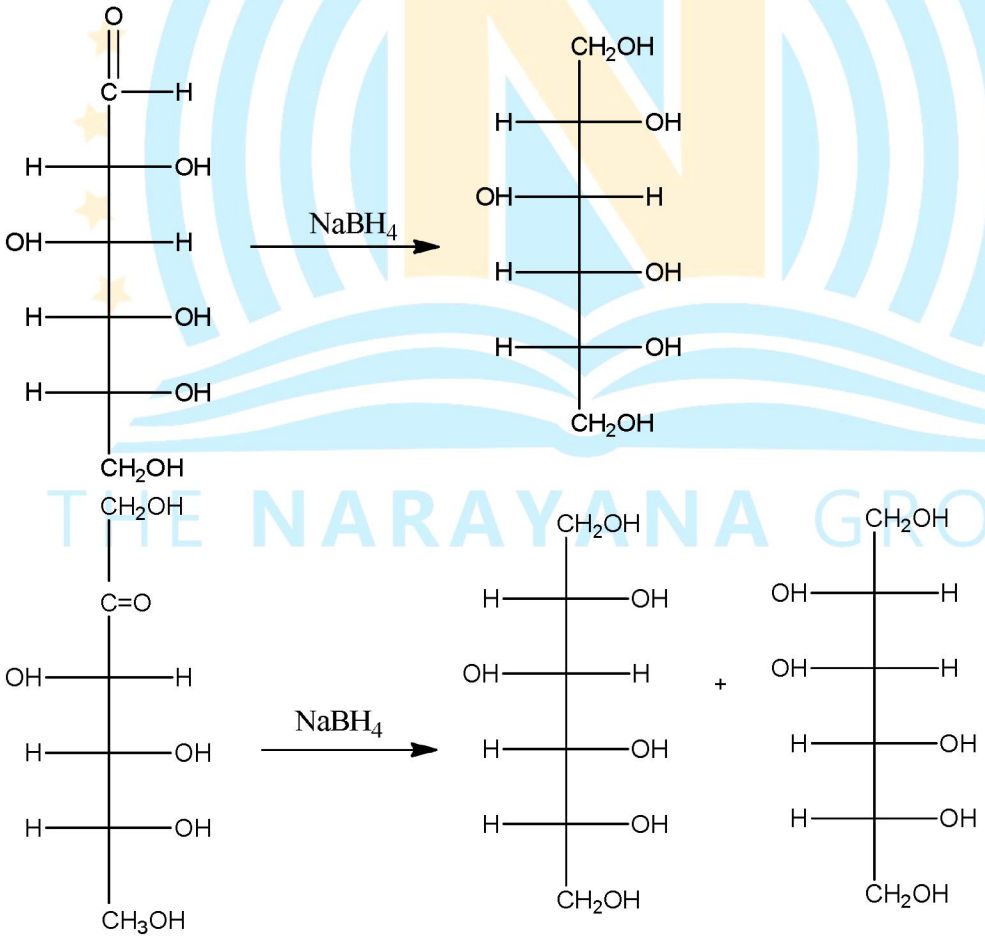


cis



optically active

49.



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50. Reaction – 1, 2, 6 is having same number of carbon atoms in reactant and final products.

MATHEMATICS

51. Let Complex number z_k correspond to point $P_k(x_k, y_k), K = 1,2$

Let P_2^1, P_2 by symmetric about the x – axis, then P_1, P_2^1 are on the right branch of the hyperbola $C: x^2 - y^2 = 2$.

Let F_1, F_2 be the foci of C , $F_1(-2,0), F_2(2,0)$ By def. of hyp_4.

$|P_1F_1| = |P_1F_2| + 2\sqrt{2}, |P_2^1F_1| = |P_2^1F_2| + 2\sqrt{2}$

$= |z_1 + 2| + |\overline{z_2} + 2| - |\overline{z_1} - z_2|$

$= |z_1 + 2| + |\overline{z_2} + 2| - |z_1 - \overline{z_2}|$

$= |P_1F_1| + |P_2^1F_1| - |P_1P_2^1|$

$= 4\sqrt{2} + |P_1F_2| + |P_2^1F_2| - |P_1P_2^1|$

$= 4\sqrt{2}$.

52. Since the 5 – digit number should be a multiple of 6.

$\therefore$ Ones place digit is either 2 or 6

Also, the sum of the digits is divisible by 3.

Also, sum of digits = 1+ 2 + 3 + 5 + 6 + 7 = 24

If ones digit is 2, then rest 4 digits should be 1, 5, 6, 7 or 1, 3, 5, 7

So, number of ways in this case = $4! \times 2$

If ones digit is 6, then rest 4 digits should be 1, 2, 5,7

So, number of ways = 4! $\therefore$ Total no. of ways = $4! \times 3 = 72$

53. $7 + 4i = (z + \overline{w})(\overline{z} - w) = |z|^2 - |w|^2 - (zw - \overline{zw})$

$|z|^2, |w|^2$ are real and Re $(zw - \overline{zw}) = 0$

then $|z|^2 - |w|^2 = 7$ and $zw - \overline{zw} = -4i$

$|z| = 3, \ |w|^2 = 2$

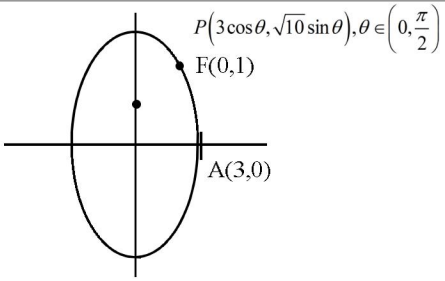
$(z + 2\overline{w})(\overline{z} - 2w) = |z|^2 - 4|w|^2 - 2(zw - \overline{zw})$

$= 9 - 8 + 8i = 1 + 8i$

Modulus of $(z + 2\overline{w})(\overline{z} - 2w) = \sqrt{65}$.

54.

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Then, $S_{OAPF}=S_{\Delta OAP}+S_{\Delta OFP}=\frac{1}{2}.3.\sqrt{10}\sin\theta+\frac{1}{2}.1.3\cos\theta$

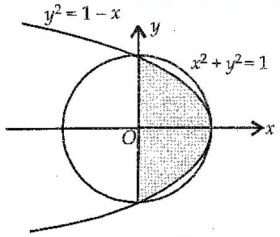
$=\frac{3}{2}\left(\sqrt{10}\,\text{Sin}\,\theta+\cos\theta\right)=\frac{3\sqrt{11}}{2}\sin\left(\theta+\phi\right)$

$\phi=\text{Tan}^{-1}\left(\frac{1}{\sqrt{10}}\right)$

When $\theta=\text{arc}\,\text{Tan}\sqrt{10}$

$\therefore$ quadrilateral OAPF has Maximum area as $\frac{3\sqrt{11}}{2}$.

55. Area $=\frac{\pi}{2}+2\int_0^1\sqrt{1-x}\,dx$



$=\frac{\pi}{2}-2.\left.\frac{(1-x)^{3/2}}{3/2}\right|_0^1=\frac{\pi}{2}-\frac{4}{3}\left[(1-x)^{3/2}\right]_0^1=\frac{\pi}{2}+\frac{4}{3}$

56. $\frac{dy}{dx}+\frac{2}{x}y=x$

I.F $=e^{\int\frac{2}{x}dx}=x^2$

$y.x^2=\int x^3dx\Rightarrow yx^2=\frac{x^4}{4}+c$

$y(1)=1$

$c=\frac{3}{4}$

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$$y=\frac{x^2}{4}+\frac{3}{4x^2}$$

57. $$\int\frac{x^2\left(x^6+x^5-1\right)dx}{\left(2x^6+3x^5+2\right)^2}$$

$$=\int\frac{x^2\left(x^6+x^5-1\right)dx}{x^6\left(2x^3+3x^2+\frac{2}{x^3}\right)^2}$$

$$=\int\frac{\left(x^2+x-x^{-4}\right)dx}{\left(2x^3+3x^2+\frac{2}{x^3}\right)^2}$$

$$2x^3+3x^2+2x^{-3}=t$$

$$6\left(x^2+x-\frac{1}{x^4}\right)dx=dt=\frac{-1}{6\left(2x^3+3x^2+2x^{-3}\right)}+c$$

58. Given, $\frac{1+2i\sin\theta}{1-i\sin\theta}$ is purely imaginary, then its real part must be zero.

$$\frac{1+2i\sin\theta}{1-i\sin\theta}\times\frac{1+i\sin\theta}{1+i\sin\theta}=\frac{\left(1+i\sin\theta+2i\sin\theta-2\sin^2\theta\right)}{1+\sin^2\theta}$$

$$=\frac{1-2\sin^2\theta}{1+\sin^2\theta}+\frac{3\sin\theta}{1+\sin^2\theta}i$$

Since real part is $0\Rightarrow\frac{1-2\sin^2\theta}{1+\sin^2\theta}=0\Rightarrow 2\sin^2\theta=1$

$$\Rightarrow\sin\theta=\pm\frac{1}{\sqrt{2}}\Rightarrow\theta=\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$$

Since, $\theta\in(0,2\pi)$

Then the sum of the elements in A is

$$\frac{\pi}{4}+\frac{3\pi}{4}+\frac{5\pi}{4}+\frac{7\pi}{4}=\frac{16\pi}{4}=4\pi$$

59. We have given $x^2+\alpha x+\beta>0$ $\therefore D<0$

$$\alpha^2-4\beta<0\Rightarrow\alpha^2<4\beta$$

$$(\alpha,\beta)=\{(1,1),(1,2),....,(1,6),(2,2),(2,3),(2,4),(2,5),(2,6),$$

$$(3,3),(3,4),(3,5),(3,6),(4,5),(4,6)\}$$

$\therefore$ Required probability $=\frac{17}{36}$

60. Let α and β be the roots of the quadratic equation

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$7x^2-3x-2=0 \quad \therefore \alpha+\beta=\frac{3}{7},\alpha\beta=\frac{-2}{7}$

Now, $\frac{\alpha}{1-\alpha^2}+\frac{\beta}{1-\beta^2}=\frac{\alpha-\alpha\beta(\alpha+\beta)+\beta}{1-(\alpha^2-\beta^2)+(\alpha\beta)^2}$

$=\frac{(\alpha+\beta)-\alpha\beta(\alpha+\beta)}{1-(\alpha+\beta)^2+2\alpha\beta+(\alpha\beta)^2}=\frac{\frac{3}{7}+\frac{2}{7}\times\frac{3}{7}}{1-\frac{9}{49}+2\times\frac{-2}{7}+\frac{4}{49}}=\frac{27}{16}$

61. Common series S with terms 11, 31 …

$T_8=11+(8-1)\times 20=11+140=151$

62. $I=\int_{-\pi}^{0}\frac{\cos^2 x}{1+a^x}dx+\int_0^{\pi}\frac{\cos^2 x}{1+a^x}dx$

$=\int_0^{\pi}\frac{\cos^2 x}{1+a^{-x}}dx+\int_0^{\pi}\frac{\cos^2 x}{1+a^x}dx$

$=\int_0^{\pi}\cos^2 x\,dx$

$=2\int_0^{\pi/2}\cos^2 x\,dx$

$=2.\frac{1}{2}.\frac{\pi}{2}=\frac{\pi}{2}$

63. Since, $7^{103}=7.(49)^{51}$

$=7(51-2)^{51}=7(51K-2^{51})$

$=17m-7.2^{51}=17m-56(17-1)^{12}$

$=17m-56(17\lambda+1)=17n-51+12-17$

$\Rightarrow$ Remainder = 12

64. General term,

$T_{r+1}={}^{10}C_r(tx^{1/5})^{10-r}\left(\frac{(1-x)^{1/10}}{t}\right)^r$

Since, it is independent of t.

$\therefore\ 10-r-r=0\Rightarrow r=5\ \therefore T_6={}^{10}C_5x(1-x)^{1/2}$

Now, $\frac{d}{dx}(T_6)={}^{10}C_5\left[(1-x)^{1/2}-\frac{x}{2\sqrt{1-x}}\right]=0$

$\Rightarrow 2(1-x)-x=0\Rightarrow 2-3x=0\Rightarrow x=2/3$

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$\therefore$ Maximum value of $T_6 = {}^{10}C_5\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)^{1/2} = \frac{2\times 10!}{3\sqrt{3}(5!)^2}$

65. Number of ways of selection of three elements in A such that $f(x) = y_2$ is ${}^7C_3$ Now for remaining 4 elements in A, we have 2 elements in B

$\therefore$ Total number of onto functions

$= {}^7C_3 \times \left(2^4 - {}^2C_1(2-1)^4\right) = {}^7C_3 \times 14$

66. Given differentiate equation is

$$\frac{dy}{dx} - \frac{3x^5\tan^{-1}(x^3)}{(1+x^6)^{3/2}}y = 2x\exp\left\{\frac{x^3 - \tan^{-1}x^3}{\sqrt{(1+x^6)}}\right\},$$

Which is in the form $\frac{dy}{dx} + Py = Q$

Here, $P = -\frac{3x^5\tan^{-1}(x^3)}{(1+x^6)^{3/2}}, Q = 2x\exp\left\{\frac{x^3 - \tan^{-1}x^3}{\sqrt{1+x^6}}\right\}$

Let $I = -\int\frac{3x^5\tan^{-1}(x^3)}{(1+x^6)^{3/2}}dx$

Put $x^3 = \tan\theta \quad \Rightarrow 3x^2dx = \sec^2\theta\, d\theta$

$$I = -\int\frac{\tan\theta.\theta}{\sec^3\theta}\times\sec^2\theta\, d\theta = -\int\theta.\sin\theta\, d\theta$$

$$= -\left[\theta\int\sin\theta\, d\theta - \int\left(\int\sin\theta\, d\theta\right)d\theta\right]$$

$$= -\left[-\theta.\cos\theta + \sin\theta\right] = -\left[\frac{x^3}{\sqrt{1+x^6}} - \tan^{-1}(x^3).\frac{1}{\sqrt{1+x^6}}\right]$$

$\therefore \quad I.F = e^{\int pdx} = e^I$

Solution of the D.E. is

$$y.e^I = \int e^I.2x.\exp\left\{\frac{x^3 - \tan^{-1}(x^3)}{\sqrt{(1+x^6)}}\right\}dx$$

$$\int 2x.e^{\frac{-x^3+\tan^{-1}(x^3)}{\sqrt{1+x^6}}}.e^{\frac{x^3-\tan^{-1}(x^3)}{\sqrt{1+x^6}}}dx$$

$= \int 2x.dx = x^2 + x \qquad \therefore \qquad ye^I = x^2 + c$

$$\therefore \qquad y = x^2\exp\left\{\frac{-\tan^{-1}(x^3) + x^3}{\sqrt{1+x^6}}\right\} + c\exp\left\{\frac{x^3 - \tan^{-1}(x^3)}{\sqrt{1+x^6}}\right\}$$

This curve passes through the origin.

APEX JEE-MAIN 2025

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So, $0=0+c \quad \Rightarrow \quad c=0$

The required solution is $y(x)=x^2\exp\left\{\dfrac{x^3-\tan^{-1}(x^3)}{\sqrt{1+x^6}}\right\}$

At x = 1 $y(1)=\exp\left(\dfrac{1-\frac{\pi}{4}}{\sqrt{2}}\right)=\exp\left(\dfrac{4-\pi}{4\sqrt{2}}\right)$

67. $(2K,3K)$ will lie on circle whose diameter is AB Equation of circle

$(x-1)(x)+(y-1)(y)=0$

$x^2+y^2-x-y=0$ ……..(i)

Satisfy $(2K,3K)$in(i)

$(2k)^2+(3k)^2-2k-k=0$

$\Rightarrow 13k^2-5k=0$

$k=0,k=\dfrac{5}{13}$

68. Let intersection points be $P(x_1,y_1)$and$Q(x_2,y_2)$ The given equations

$x^2=4y$ …….(i)

$x-\sqrt{2}y+4\sqrt{2}=0$ ……(ii)

Use eqn (i) eqn (ii)

$x-\sqrt{2}\dfrac{x^2}{4}+4\sqrt{2}=0, \qquad \sqrt{2}x^2-4x-16\sqrt{2}=0$

$x_1+x_2=2\sqrt{2},x_1x_2=-16,(x_1-x_2)^2=8+64=72$

Since, points P and Q both satisfy the equations (ii), then

$x_1-\sqrt{2}y_1+4\sqrt{2}=0, \;\; x_2-\sqrt{2}y_2+4\sqrt{2}=0$

$(x_2-x_1)=\sqrt{2}(y_2-y_1)\Rightarrow(x_2-x_1)^2=2(y_2-y_1)^2$

$\Rightarrow PQ=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$=\sqrt{(x_2-x_1)^2+\dfrac{(x_2-x_1)^2}{2}}$

$=|x_2-x_1|.\dfrac{\sqrt{3}}{\sqrt{2}}=6\sqrt{2}\times\dfrac{\sqrt{3}}{\sqrt{2}}=6\sqrt{3}$

Hence, length of chord = $6\sqrt{3}$.

70. $\int(xe^{\tan x}+xe^{\tan x}(1+x\sec^2 x)dx$

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$$\int\left(f(x)+xf^{1}(x)\right)dx = xf(x)+C$$

$$= x^{2}e^{\tan x}+C$$

71. $$29\left(\frac{31}{x+2}-7+\frac{31}{x+3}+8\right)=31\left(\frac{29}{x+1}-5+\frac{29}{x+4}-6\right)+370$$

$$29\times 31\left(\frac{1}{x+2}+\frac{1}{x+3}\right)=31\times 29\left(\frac{1}{x+1}+\frac{1}{x+4}\right)$$

$$\frac{1}{x+2}+\frac{1}{x+3}=\frac{1}{x+1}+\frac{1}{x+4}$$

$$x^{2}+4x+3=x^{2}+6x+8$$

$$\Rightarrow x=-5/2 \qquad \text{Ans: } 5+2=7.$$

72. $$I=\int\frac{(e^{x}-1)(\sin x-\cos x)+x\cos x}{\sin^{2}x\left\{1+\left(\frac{e^{x}-1-x}{\sin x}\right)^{2}\right\}}dx$$

Let $\frac{e^{x}-1-x}{\sin x}=t,$

$$I=\int\frac{1}{1+t^{2}}dt=\tan^{-1}\left(\frac{e^{x}-1-x}{\sin x}\right)+c$$

$$f(x)=\frac{e^{x}-1-x}{\sin x}$$

$$\lim_{x\to 0}\frac{f(x)}{x}=\lim_{x\to 0}\frac{e^{x}-1-x}{x\left(\frac{\sin x}{x}\right)}=\frac{1}{2}=\frac{p}{q}$$

$$\frac{(p+q)}{\text{least}}=3.$$

73. $2^{N}<N!$, which is true when $N\geq 4$

N = 1 (Not possible)

N = 2 i.e., (1, 1) (Not possible)

N = 3 i.e., (1, 2) (2, 1) (Not possible)

$\therefore$ Required probability $=\frac{36-3}{36}=\frac{33}{36}=\frac{11}{12}$

$\therefore$ m = 11 and n = 12

Now, $4m-3n=4(11)-3(12)=44-36=8$

74. Points satisfying the inequalities are common to all 3 circles, given by (1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (3, 2),

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75. $A_n = 704\dfrac{\left(1-\left(\dfrac{1}{2}\right)^n\right)}{1-\dfrac{1}{2}} = 1408\left(1-\left(\dfrac{1}{2}\right)^n\right).$

$B_n = (1984)\left(\dfrac{2}{3}\right)\left(1-\left(\dfrac{-1}{2}\right)^n\right)$

$A_n = B_n$

$33\left(1-\left(\dfrac{1}{2}\right)^n\right) = 31\left(1-\left(\dfrac{-1}{2}\right)^n\right)$

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